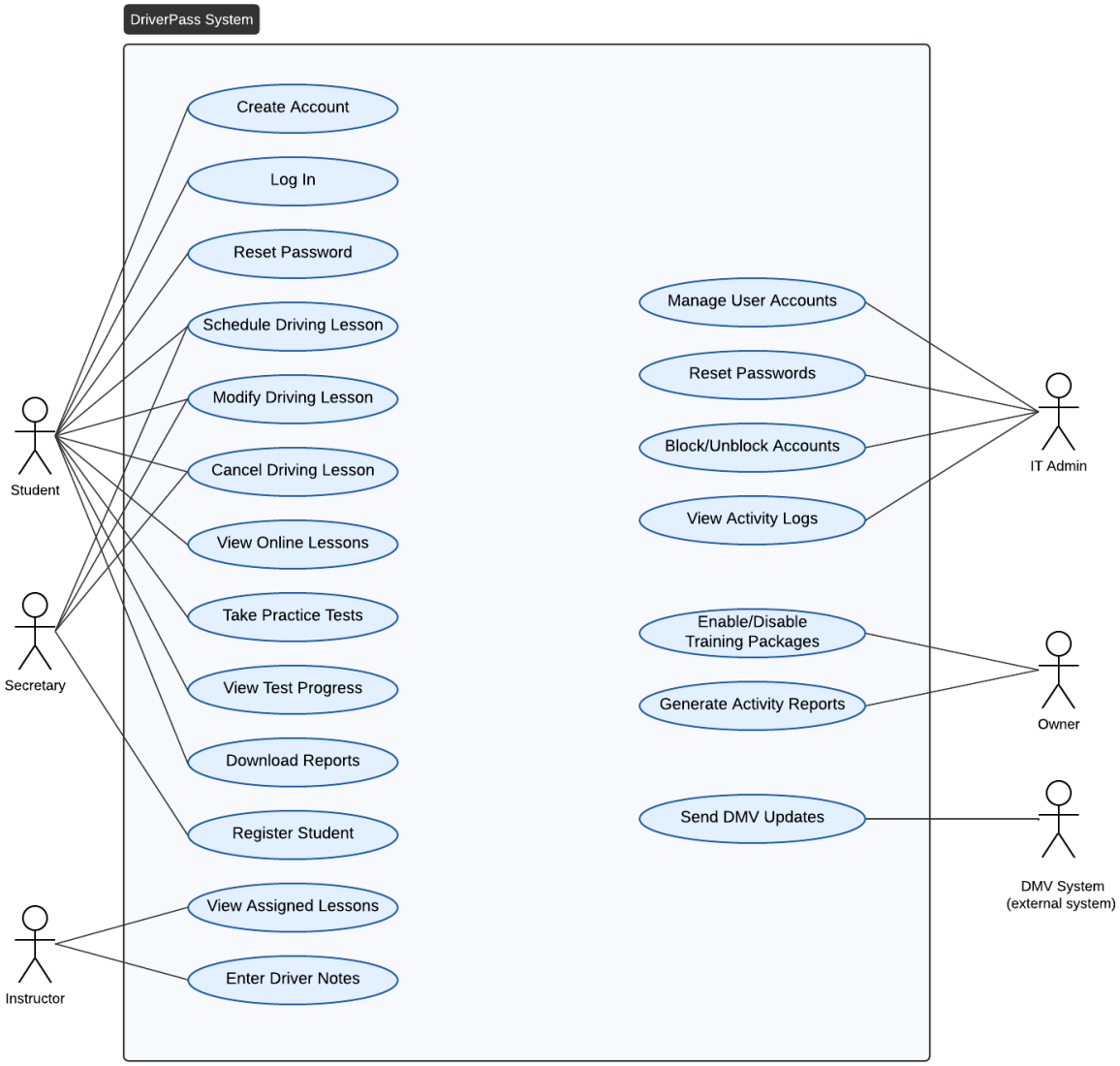
CS 255 System Design Document Template

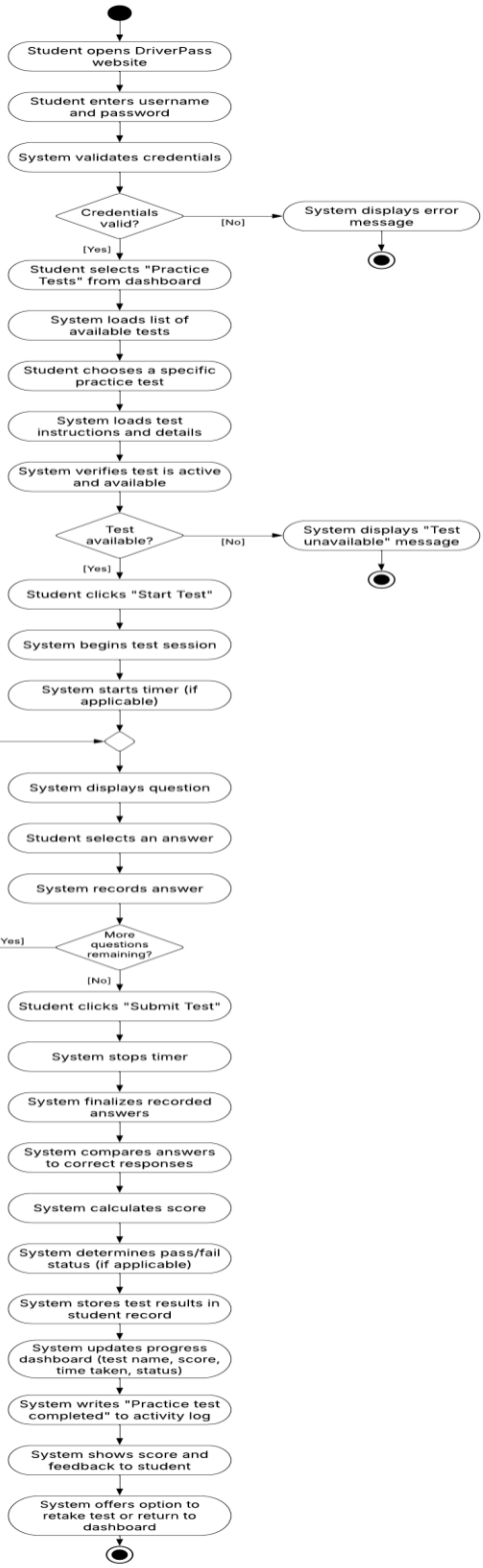
UML Diagrams

UML Use Case Diagram

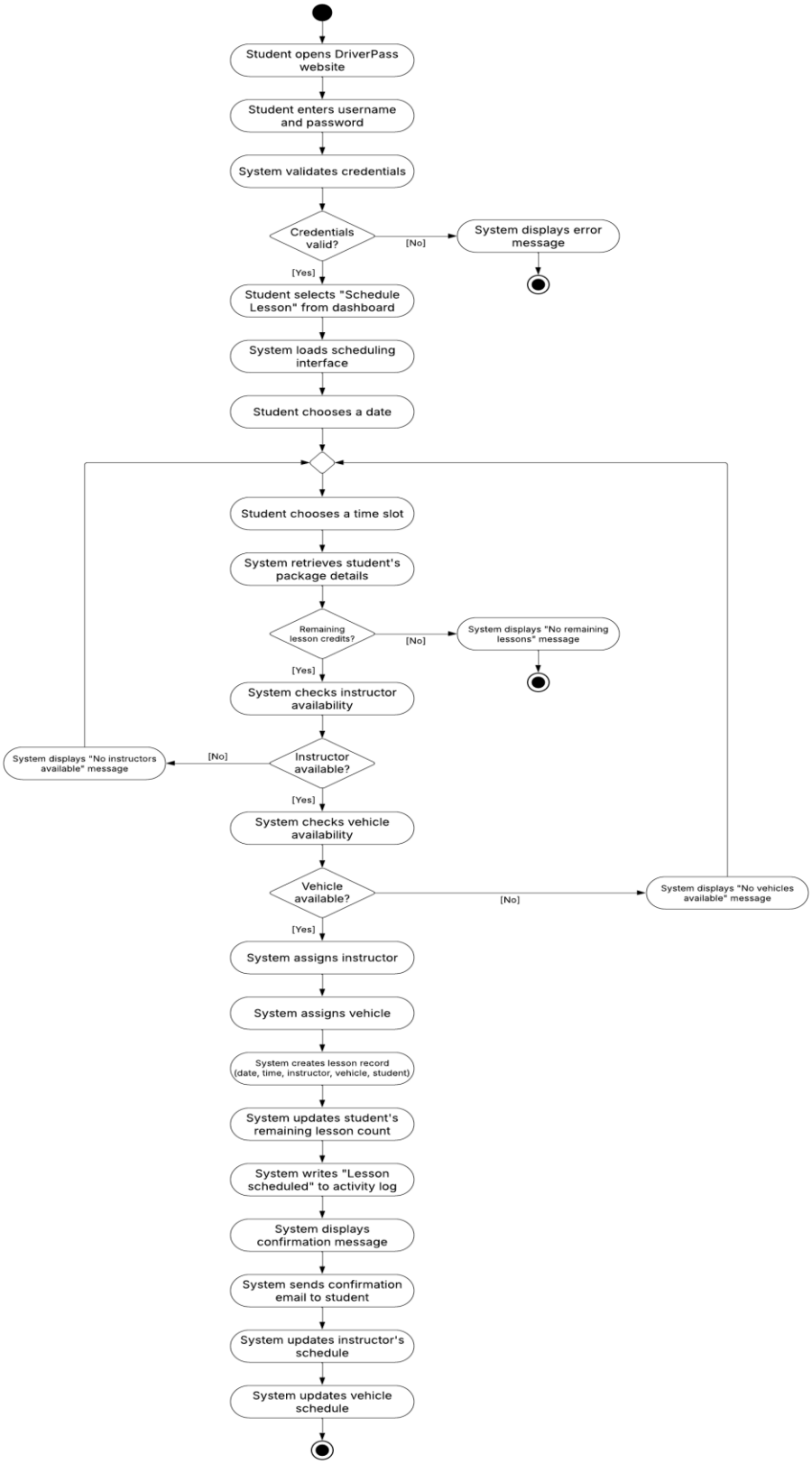


UML Activity Diagrams

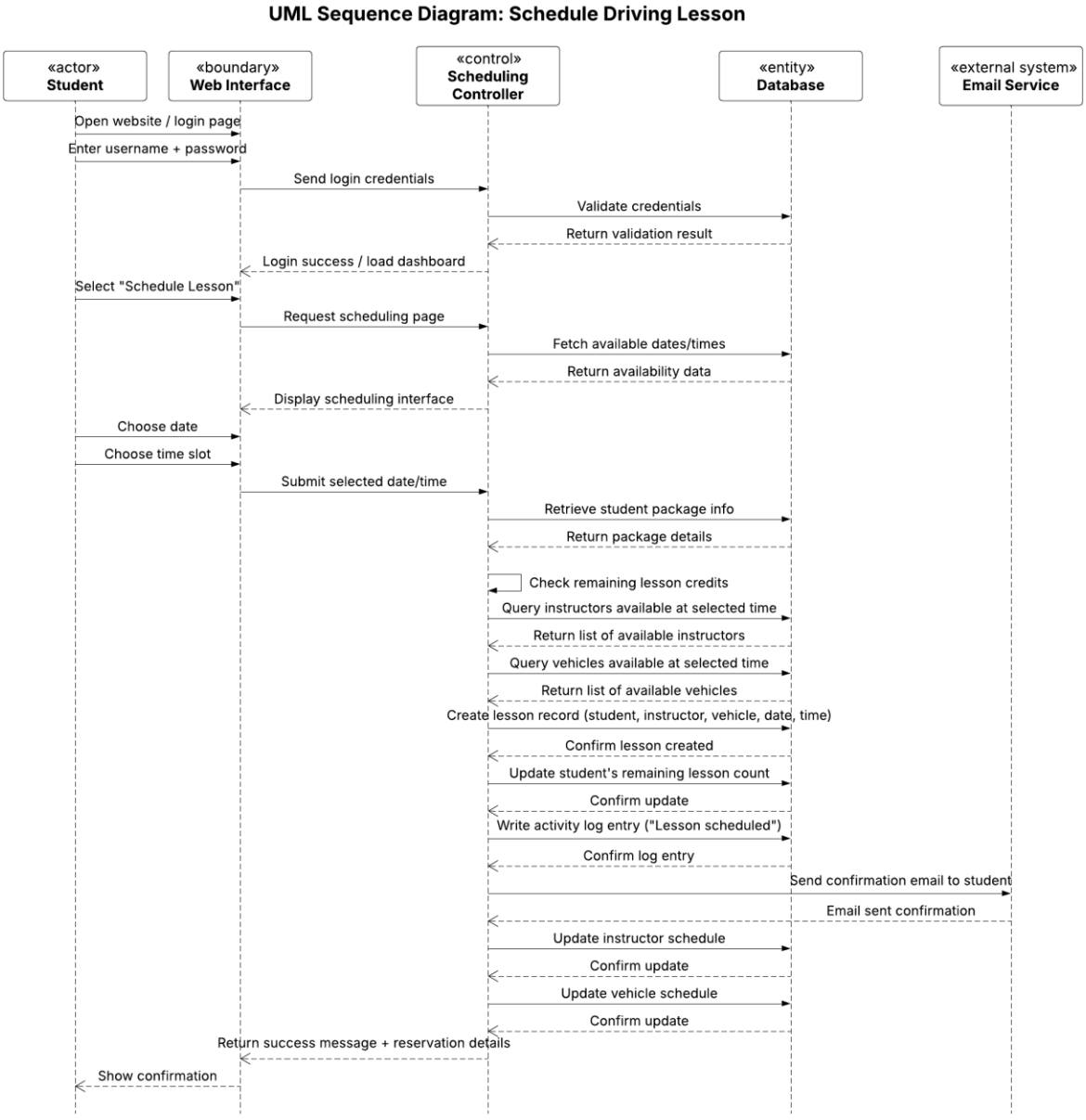
UML Activity Diagram: Take Online Practice Test



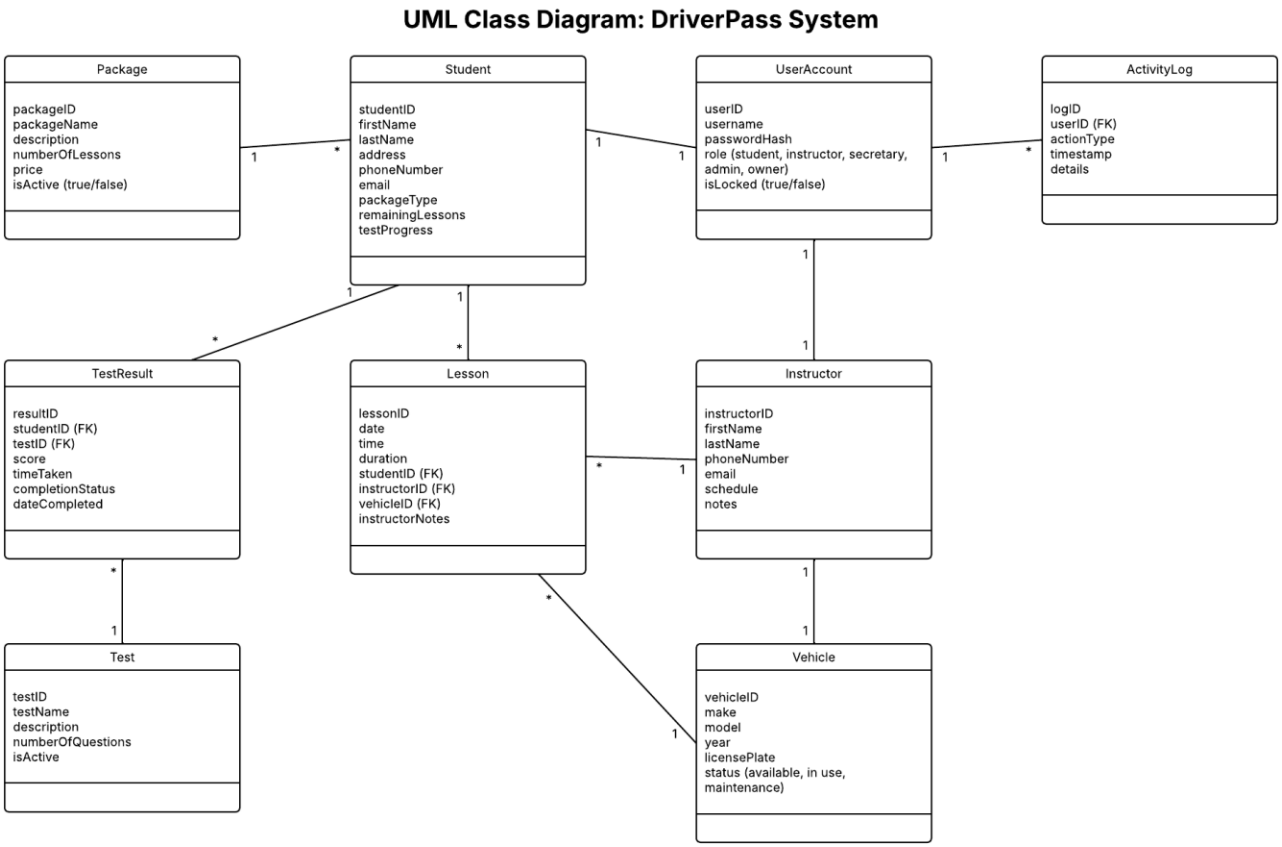
UML Activity Diagram: Schedule Driving Lesson



UML Sequence Diagram



UML Class Diagram



Technical Requirements

Hardware
The DriverPass system doesn't need any special hardware. It just needs to run on the devices people already use every day like computers, tablets, and phones. As long as the device has a modern web browser, the system should work without any extra installation. Everything will be hosted in the cloud, which means DriverPass doesn't have to maintain physical servers or deal with storage.

Software
The system will run as a browser-based web application. The server will handle things like logging in, scheduling lessons, storing progress, and managing user roles. A secure cloud database is required to keep track of accounts, lesson schedules, instructor information, test results, and activity logs. The system also needs a way to pull in DMV updates, so new rules or test questions can be added without major changes.

Tools
The development team will use standard web technologies and a cloud provider for hosting. Lucidchart is used during the design phase to build the UML diagrams, and the admin side of the system will include built-in tools for managing users, packages, and system settings. Staff should be able to download reports (like Excel files) without needing any extra software.

Infrastructure
The infrastructure needs to support secure communication, so all data between the user and the server should be encrypted. Role-based access is required, so students, instructors, admins, and the owner only see what applies to them. The system should lock accounts after repeated failed login attempts and support password resets through email. Regular backups, activity logs, and monitoring tools are also part of the setup, so the IT officer can keep the system running smoothly. Because everything is based in a browser, the system should keep working even when browsers update or new devices come out.